



AYRTON SENNA

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Jakarta, Indonesia

Mechanical Engineering graduate from Universitas Pertamina (GPA 3.79/4.00) with hands-on experience in mechanical engineering, reliability, maintenance, and project engineering. Experienced in mechanical equipment, piping systems, firewater systems, engineering design validation, and technical document review through internship and projects. Demonstrated analytical and problem-solving capabilities through maintenance analysis, FMEA, engineering assessments, modelling, and technical evaluations. Strong leadership, teamwork, communication, and learning agility developed through multidisciplinary engineering projects, industrial internships, research, and organizational leadership.

Experiences

PT Abimantrana Engineering Technology - Jakarta

Mar 2026 - Jul 2026

RnD Engineer (Project Based)

- Engineered the prototype of an automated cocktail machine by improving system thermal efficiency.
- Designed and integrated inline carbonation systems, internal stirring mechanisms, and precision piping assemblies to achieve consistent beverage mixing and controlled fluid delivery.
- Optimized mechanical system performance through component integration and prototype testing, enhancing operational reliability and manufacturing feasibility.
- Collaborated with a multidisciplinary engineering team to translate product concepts into a functional prototype, supporting rapid product development and technical validation.

PT Pertamina Maintenance & Construction (PertaMC) - Jakarta

Aug 2025 - Oct 2025

Mechanical Engineer Intern

- Reviewed multiple engineering documents including P&IDs, Equipment Datasheets, and Line Class specifications to support the Firewater Pump House Addition Project at RU IV Cilacap.
- Verified pump configurations, piping material specifications, and safety devices (PSV and Check Valves) against NFPA 20 and ASME B31.3 requirements.
- Supported engineering design validation to improve mechanical system reliability and maintain project compliance.

PT Abimantrana Engineering Technology - Jakarta, Indonesia

Apr 2025 - Jul 2025

RnD Engineer (Project Based)

- Contributed to the Basic Engineering Design (BED) and Front-End Engineering Design (FEED) of HelionPod X, an integrated energy system combining Concentrated Solar Power (CSP), Stirling Engine, and hydrogen storage technologies.
- Prepared conceptual mechanical layouts and integrated subsystems to ensure compatibility, maintainability, and overall system functionality during the early engineering design phase.
- Performed preliminary engineering calculations and technical evaluations to assess system performance, design feasibility, and energy efficiency.
- Supported engineering design reviews through risk identification, safety assessments, and design verification, improving project reliability and supporting informed engineering decisions.

PT Pertamina Patra Niaga

Jul 2025 - Oct 2025

Project Assistant

- Assisted in a vapour dispersion modelling project to support the implementation of gas and flame detectors at fuel terminal tank areas.
- Preparing and validating input data using Excel
- Developing dispersion scenarios in ALOHA and Google Earth Pro software for 100+ tank in the SUMBAGUT & KALIMANTAN areas.
- Assisting in report preparation for client deliverables

Mechanical Engineering, Universitas Pertamina - Jakarta, Indonesia

Mar 2025 - Jul 2025

Practicum Assistant for Fluid Mechanics Laboratory

- Directed hands-on experiments for Mechanical Engineering students on pipe flow systems, pump operations, and hydraulic measurements.
- Enforcing strict safety compliance.
- Enhanced technical comprehension of fluid dynamics principles and data interpretation through real-time troubleshooting guidance.
- Developed students' industry-relevant skills in experimental analysis for energy sector applications.

PT Astra Honda Motor - Karawang, Indonesia

Jan 2025 - Feb 2025

Maintenance Engineer Intern

- Preventive Maintenance: Conducting regular inspections and machine maintenance to prevent potential operational disruptions.
- Corrective Maintenance: Assisting in repairs when machine failures or malfunctions occurred.
- Reporting Worksheet: Supporting the documentation of inspection results and maintenance activities.
- Failure Analysis using FMEA: Performing failure analysis on the flow process using the Failure Mode and Effects Analysis (FMEA) method in the cylcomp line to identify potential risks and enhance process reliability.

Mechanical Engineering, Universitas Pertamina - Jakarta, Indonesia

Sep 2024 - Jan 2025

Practicum Assistant for Strength of Materials and Material Engineering Laboratory

- Guided students through hands-on experiments on material properties, stress-strain analysis, and mechanical testing.
- Ensuring safety protocol adherence and comprehension of experimental techniques.
- Enhanced students' grasp of core concepts and technical communication skills through clear instruction and a collaborative approach in an interactive learning environment.

Universitas Pertamina - Jakarta, Indonesia

Jul 2024 - Sep 2024

Research Assistant in BIO Metal-Organic Frameworks (MOF) for Carbon Capture

- Developed and characterized sustainable Bio-MOF materials for efficient carbon capture from industrial emissions.
- Optimized material properties through synthesis, gas adsorption experiments, and performance analysis.
- Collaborated with cross- functional teams to advance CO2 capture efficiency. Contributed to research papers and technical presentations documenting findings.

Colleges

Universitas Pertamina - Jakarta Selatan

Sep 2022 - Aug 2026

Bachelor of Mechanical Engineering, 3.79/4.00

- Final Thesis: Analisis Kejadian Overfill pada Tangki Penyimpanan BBM Terhadap Pembentukan Zona Bahaya dan Risiko Efek Domino Akibat Dispersi Uap dan Synergistic Effect of Explosion and Fire (SEEF): Studi Kasus CAPECO
- Beasiswa Unggulan Semester (BUS) 2023 & 2026
- Beasiswa Haji BPKH 2024
- Publication (Motivection Publisher): A Comparative Study of Human Reliability Assessment using Success Likelihood Index Method (SLIM) and Human Error Assessment & Reduction Technique (HEART): A case study from a Boeing 737 Max Accident

SMAN 2 PURWAKARTA - Purwakarta

Jun 2019 - Apr 2022

Organisational & Volunteer Experience

AMPP SC UPer - Jakarta

Jul 2025 - May 2026

Chairman

- Led and coordinated the core team to achieve the chapter's organizational vision.
- Developed and oversaw webinars, workshops, and industry visits aligned with AMPP standards.
- Built networks with industry professionals and academics to strengthen member development opportunities.
- Drove member engagement through initiatives supporting innovation and professional development.

LIGHT UP 2024 - Barengkok Village, Bogor

Mar 2024 - Aug 2024

Event Division

LIGHT UP is a community service institution under the auspices of the Director of Student Affairs at Pertamina University.

- Contributed to the strategic planning and on-ground logistics for community activities, ensuring smooth operations.
- Coordinated directly with village representatives and community members to tailor programs to local needs.
- Managed event timelines, resource allocation, and participant coordination for scheduled activities.
- Gained hands-on experience in project management within a social context, developing strong problem-solving and cross-cultural communication skills.

Abimantrana - Designing and developing a car in the Gasoline Urban Concept category to compete in the Kontes Mobil Hemat Energi (KMHE) organized by Puspresnas.

Dec 2022 - May 2023

General Manager

Abimantrana is an organization focused on the fields of automotive and clean energy.

- Concept Development ; Research and analyze design concepts for energy-efficient vehicles in the Gasoline Urban Concept category, Create innovative ideas to optimize vehicle performance, fuel efficiency, and sustainability.
- Design and Engineering ; Develop 3D models and detailed designs using CAD software, Conduct structural and aerodynamic simulations to ensure safety and efficiency, Collaborate with the team to integrate mechanical, electrical, and control systems.

Dewan Perwakilan Angkatan Teknik Mesin Universitas Pertamina - Jakarta

Sep 2022 - Mar 2023

Anggota

Served as an elected representative on the student legislative body responsible for the foundational governance of the student association. My primary responsibility involved a comprehensive review and reform of the Association's Constitution and Bylaws (Anggaran Dasar/ Anggaran Rumah Tangga - AD/ART).

- Collaborated with fellow council members to analyze existing articles, identify areas for improvement, and draft proposed amendments.
- Facilitated discussions with the executive board and student body to gather feedback and build consensus on significant changes.
- Ensured all revisions aligned with the university's regulations and the long-term strategic goals of the association.
- Strengthened the legal framework of the organization to enhance transparency, accountability, and operational efficiency.

Lecturer Community Service - Tangerang

Oct 2023 - Dec 2023

As a Speaker in a Presentation Explaining Line-Following Robots

- Delivered a presentation on the concept and functionality of line- following robots, followed by hands-on practice guiding SMAK Tunas Bangsa students in building their own line-following robots.

Skills & Project

- **Engineering & Technical** (2026): Reliability Engineering | Maintenance | Piping Engineering | Pump Systems | P&ID Interpretation | Equipment Datasheet Review | Engineering Drawing Review | Engineering Documentation | Valve Selection | FMEA | Dispersion Modelling | ALOHA | SolidWorks | Python | Microsoft Office
- **Professional** (2026) : Critical Thinking | Problem Solving | Analytical Thinking | Leadership | Teamwork | Public Speaking | Communication | Time Management | Intermediate English
- **TEH PICAH: Innovation of an Integrated Waste Bin with Sensor-Based Automatic Sorting and Shredding Machine** (2025): Developed a prototype integrating infrared, capacitive, and inductive proximity sensors with a mechanical shredding mechanism to automate waste segregation and enhance recycling efficiency.
- **LIGHT UP 2024 " Engraving the Spirit of Independence: Together with Barengkok Village"** (2024): Coordinated educational and cultural programs for Barengkok Village, fostering community participation through collaborative planning and event execution.
- **Hero Helmets: An Innovative Utilization of Water Hyacinth Fiber for Safety Helmet Production in the Mining Industry** (2024): Designed a lightweight mining safety helmet utilizing water hyacinth fiber composites, supporting sustainable material innovation while improving structural performance and user comfort.